

Key Drivers of Employee Attrition

An analysis of the IBM HR Analytics Employee Attrition dataset

Overview

This analysis examines 1,470 employee records from the IBM HR Analytics dataset to identify the strongest predictors of voluntary employee attrition. Using pivot analysis across department, overtime status, and job satisfaction, three consistent and actionable drivers of turnover were identified. The findings below are intended to mirror the type of rapid diagnostic a People Advisory or Human Capital consulting team would produce for a client in an early-stage retention review.

Key Findings

31% vs 10% Overtime vs. no overtime	23% vs 11% Lowest vs highest satisfaction	21% Highest department (Sales)
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- Overtime is the strongest single predictor of attrition.** Employees who worked overtime left at a rate of 31%, compared with 10% for those who did not — roughly three times higher. This points to workload and burnout as a likely primary driver of voluntary turnover.
- Attrition falls steadily as job satisfaction rises.** Attrition declines from 23% among the least satisfied employees to 11% among the most satisfied, a consistent gradient rather than a one-off gap — indicating satisfaction is a reliable, not coincidental, predictor.
- Attrition is concentrated in Sales, not spread evenly across the business.** Sales shows the highest departmental attrition at 21%, ahead of HR (19%) and R&D (14%) — notable given R&D is the largest department by headcount. This suggests the issue is localised rather than organisation-wide.

Recommendations

- Prioritise a workload and overtime review within the Sales function, where both overtime prevalence and attrition risk appear to intersect.
- Investigate root causes of low job satisfaction through targeted pulse surveys or exit interviews, focused on the lowest-scoring segment.
- Pilot a retention intervention (e.g. workload rebalancing or flexible scheduling) for employees currently logging regular overtime, then track attrition over the following two quarters.

Methodology: Pivot table analysis (Department × Attrition, OverTime × Attrition, JobSatisfaction × Attrition) performed in Excel on the IBM HR Analytics Employee Attrition dataset (Kaggle, 1,470 records, 35 variables).